Post-hoc Concept Refinement Improves Performance in Vision Foundation Models

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Abstract

Foundation models trained on large, diverse datasets learn rich representations that achieve state-of-the-art performance across numerous tasks. However, we hypothesize that this generality comes at a cost: the resulting embeddings encode a superposition of many concepts, only a subset of which are relevant for any specific downstream input. We propose a post-hoc refinement method that leverages sparse autoencoders (SAEs) to isolate salient concept directions within pre-trained embeddings. By training an SAE to learn a sparse, overcomplete basis of latent concepts and reconstructing embeddings using only the top-K most active features, we generate refined representations that retain task-relevant concepts while suppressing irrelevant signals. We validate this approach on OpenMidnight, a histopathology foundation model, across five benchmarks, and on BrainIAC, a brain MRI foundation model, across four downstream tasks. The Results demonstrate improvements over baseline linear probing, positioning the concept-directed refinement as a generalizable strategy for enhancing foundation model transfer performance.

1 Introduction

Foundation models trained on large-scale medical imaging datasets have demonstrated remarkable transferability across diverse downstream tasks. These models learn rich representations that capture complex semantic concepts ranging from tissue architecture to subtle pathological patterns. Recent work in mechanistic interpretability has revealed that neural networks exhibit concept superposition: the condition where multiple features are represented in overlapping, non-orthogonal directions within the activation and representation space [Elhage et al., 2022]. Foundation models exhibit this property to the fullest extent, allowing them to encode numerous concepts within a fixed-dimensional embedding space, thereby enabling broad applicability and generalizability. However, we hypothesize that this property also imposes constraints on downstream performance (e.g., zero-shot testing and linear probing), since for any given input only a small subset of these encoded concepts are relevant. The presence of numerous, task-irrelevant, and superimposed feature directions could act as noise, potentially diluting the signal and degrading performance.

Sparse autoencoders (SAEs) have been used to uncover the concepts driving a model's predictions by learning to decompose model activations into a larger, sparse set of features, providing valuable pathways for interpretation [Bricken et al., 2023, Cunningham et al., 2023, Templeton et al., 2024]. However, the potential of using these disentangled concepts to directly improve a model's downstream performance remains largely unexplored.

In this work, we ask: can we leverage the sparse concept decomposition learned by an SAE to refine the output embeddings of a foundation model, creating a representation that is more performant for downstream tasks? To this end, we propose a framework that employs SAEs to disentangle and reweight concept directions within the [CLS] token embeddings of a foundation model. The key

Preprint. Under review.

insight is that by expanding the latent space and enforcing top-K sparsity, we can reconstruct a refined embedding that improves downstream performance.

The contributions of this work are threefold:

1. We demonstrate that top-K concepts in an expanded latent space can effectively reconstruct a refined, denoised embedding for foundation models.
2. We propose a monitoring metric, "Usage Fraction," to estimate representation collapse and decomposition ability.
3. We show performance gains across two distinct medical imaging modalities (histopathology and brain MRI), establishing the generalizability of post-hoc feature refinement.

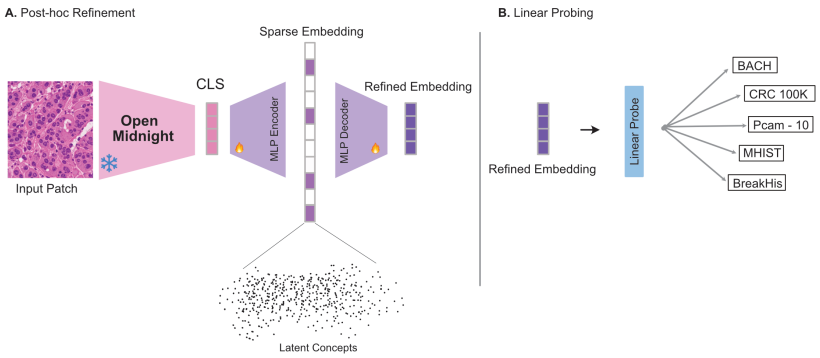

Figure 1: Overview of the proposed SAE-based embedding refinement framework. Frozen embeddings from a foundation model are passed through a sparse autoencoder. The top-K most active latent features are retained(A), and the decoder reconstructs a refined embedding used for downstream tasks (B).

2 Related Work

2.1 Mechanistic Interpretability and Sparse Autoencoders

Recent work has demonstrated that SAEs can effectively uncover and disentangle interpretable concepts from foundation models in medical imaging, providing pathways toward interpretability in healthcare AI models. Abdulaal et al. [2024] applied SAEs to decompose latent representations from a pretrained radiology model into human-interpretable features, using large language models to label discovered features and subsequently compile them into radiology reports. Concurrently, Renzulli et al. [2025] applied SAEs to MedCLIP embeddings and proposed a systematic evaluation framework combining correlation metrics, entropy analysis for monosemanticity, and automated neuron naming via MedGEMMA, demonstrating that SAE neurons exhibit stronger alignment with clinical concepts than raw model features. These studies show that SAEs can recover semantically meaningful directions from medical vision and vision-language models, providing much-needed interpretability for medical AI systems. However, while these methods focus on post-hoc interpretation of learned representations, whether the extracted concept directions can be leveraged to actively refine embeddings for improved downstream performance remains unexplored.

3 Method

3.1 Conceptual Framework

Let $x \in \mathbb{R}^d$ denote the frozen embedding from a foundation model. Following the superposition hypothesis, we assume:

$$x = \sum_{i=1}^{N} \alpha_i c_i$$

where c_i represent latent concept directions and superposition coefficients α_i correspond to the activation strength of each concept. For any given input, many coefficients α_i should be near-zero, representing inactive or weakly relevant concepts. However, greater superposition implies that multiple concepts share embedding dimensions, which during downstream probing may introduce noise.

3.2 Sparse Autoencoder

We train a sparse autoencoder (SAE) on pre-extracted embeddings pf the foundation model. The encoder maps the input embedding to a latent representation:

$$z = W_e x$$

Top-K sparsity is enforced by retaining only the K largest activations:

$$\tilde{z}_i = \begin{cases} z_i & \text{if } i \in \text{TopK}(z) \\ 0 & \text{otherwise} \end{cases}$$

The decoder reconstructs the embedding from the sparse latent code:

$$\hat{x} = W_d \tilde{z}$$

The reconstruction objective is defined as:

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|_2^2$$

The reconstructed feature $\hat{x}$ is used as the refined embedding for downstream tasks.

3.3 Latent Expansion

To increase representational capacity, the latent space is expanded [Templeton et al., 2024]:

$$\dim(z) = m \cdot d$$

with $m = 64$. Larger expansion enables more disentangled concept directions while sparsity is enforced via fixed K.

3.4 Monitoring Collapse

A potential failure mode is latent collapse, where the same K units activate for all samples. To detect latent collapse, we created a metric monitor the fraction of latent units utilized per batch:

$$\text{Usage Fraction} = \frac{\text{Number of unique active units in batch}}{\text{Total latent units}}$$

Stable increase, non-zero usage fraction indicates distributed concept learning.

4 Experimental Setup

4.1 Foundation Models

OpenMidnight (OM): A pathology vision foundation model with output embedding dimension $d = 1536$. We trained our SAE on embeddings extracted from 50% and 100% of its original pretraining dataset. The provided index files for both the 100% and 50% datasets from TCGA were used to inherit the random patch sampling performed in the original work and keep the embedding distribution consistent across experiments.

BrainIAC: A vision foundation model for brain MRI data, with output embedding dimension $d = 768$. We used 100% of the pretraining dataset for the training of the SAE.

4.2 SAE Training

For both models, we used an SAE with a latent dimension of $64 \times d$ and a top-k sparsity constraint with $k = 64$, following recommendations from Templeton et al. [2024]. We used the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 512, with early stopping. Training was conducted using the reconstruction objective with MSE as the loss function. We logged both the MSE loss and Usage Fraction to monitor for training collapse. Checkpoints were saved based on the MSE metric.

4.3 Downstream Evaluation

We evaluated the quality of the refined embeddings using linear probing. To ensure fair and consistent comparison with the baseline OpenMidnight model, we conducted all evaluations using the EVA framework, adopting the exact hyperparameters and evaluation protocol defined in the OpenMidnight configuration. For OpenMidnight, evaluation was performed on five benchmarks (MHIST, CRC-100K, PCAM-10 shots, BreakHis, BACH), with appropriate metrics reported in Table 1.

Following the same protocol, we evaluated the refined embeddings on BrainIAC using linear probing. To ensure consistency, we adopted the standard evaluation splits and hyperparameters defined in the BrainIAC repository. Evaluation was performed on four medical imaging tasks (BrainAge, Time to Stroke Onset, Mild Cognitive Impairment (MCI), IDH Mutation), with appropriate metrics reported in Table 2.

5 Results

5.1 Results on OpenMidnight

Table 1 presents the linear probing results for OpenMidnight. OM-SAE (100%) outperforms the baseline (OM) on 3 out of 5 benchmarks, with OM-SAE (50%) outperforming others on PCAM-10 shots.

Table 1: Linear probing evaluation on EVA benchmarks (↑ = higher is better).

Model	BACH (↑)	PCAM (10) (↑)	BreakHis (↑)	CRC-100K (↑)	MHIST (↑)
Midnight	0.904	0.791	0.841	0.966	0.807
OpenMidnight	0.916	0.790	0.873	0.961	0.844
OM-SAE (50%)	0.906	**0.892**	0.717	0.965	0.735
OM-SAE (100%)	**0.922**	0.804	0.753	**0.967**	0.832

5.2 Results on BrainIAC

Table 2 shows that our proposed method generalizes to a different medical imaging domain. On two out of four tasks, BrainIAC-SAE demonstrates improved performance. For the regression tasks (BrainAge and Time to Stroke), we observe minimal change, while for the classification tasks, performance improves.

Table 2: Linear probing evaluation on BrainIAC downstream tasks. (↓ = lower is better, ↑ = higher is better).

Model	BrainAge (MAE ↓)	Time to Stroke (MAE ↓)	IDH (AUC ↑)	MCI (AUC ↑)
BrainIAC	81.3	64.47	0.68	0.72
BrainIAC-SAE	81.3	64.47	**0.71**	**0.76**

5.3 Analysis of SAE Training Dynamics

As shown in Figure 2, the Usage Fraction increases and stabilizes during training, indicating that the SAE learns to utilize a diverse set of latent features rather than collapsing onto a small, redundant subset.

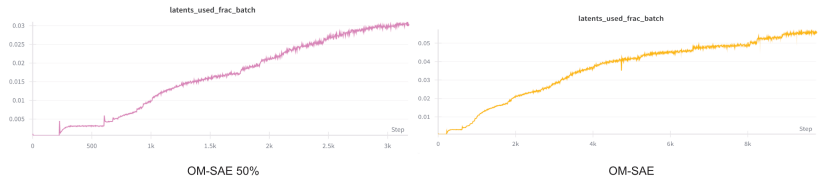

Figure 2: Usage fraction over training steps for OM-SAE 50% and OM-SAE 100%.

6 Discussion

The performance improvements across two distinct models validate the hypothesis that foundation model embeddings contain a superposition of concepts, and that isolating the most relevant ones via a sparse autoencoders can lead to better downstream performance.

6.1 Limitations and Future Work

The current approach operates on the final [CLS] token embedding. While effective for classification and regression tasks, it is not directly applicable to dense prediction tasks like segmentation, which require patch embeddings. A natural extension is to train SAEs on the internal activations of the foundation model (e.g., at various transformer blocks) such that the model outputs a refined embedding.

Furthermore, this method is post-hoc; it refines pre-existing concepts but does not enable the model to learn new ones. A promising future direction is to integrate sparsity-enforcing mechanisms directly into the pretraining objective. This could provide two major benefits: (1) it would encourage the model to learn an even larger set of more orthogonal concepts, reducing superposition from the start, and (2) it would make the concept directions more innate to the model, potentially enabling more direct forms of interpretability and control, such as "steering" the model by modulating specific concept directions (e.g., turning off an "acquisition noise" concept to improve generalization).

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